Surrogate comparison: metrics reference

Written by `scripts/surrogate_comparison/compute_metrics.py` into `metrics.yaml` (scalars per source: {`mean`, `per_sample`}) and `metrics.nc` (curves on dims `source`×`lead`, sample means). Each formula is one function in `scripts/utils/surrogate_metrics.py`; the module docstrings are the authority if this sheet and the code ever disagree.

Conventions

- **Fields.** $u_c(\mathbf{x}, t_n)$, $c \in \{u, v, w\}$ the truth (LES), $\hat{u}_c$ a prediction; $n = 1..N$ the lead steps. $\mathcal{F}$: the fluid cells; $\mathcal{I} \subset \mathcal{F}$: the fluid cells whose six face neighbours are fluid (not on the domain edge), where the central differences $(f_{i+1} - f_{i-1})/2h$ of vorticity and divergence read only fluid values.
- **Window.** Lead steps $1..N$, N the shortest rollout of all models and samples (the start frame, lead 0, is not in it).
- **Two means.** $\bar{u}_c(\mathbf{x}) = \frac{1}{N}\sum_n u_c(\mathbf{x}, t_n)$, the *truth's* per-cell time mean over the window, is the one reference of every per-lead fluctuation (ACC, spectra, Λ_y, PDFs, W_2): the same for every source, so a model is never rewarded for its own drift. The *window statistics* (TKE curve and fields, profiles) use each source's *own* time mean $\langle\cdot\rangle_t$ over the window. Both depend on the window length.
- **Reference scales** (per sample, from the truth window; in `metrics.yaml` `reference` and `metrics.nc` `reference_*`, `canopy_level`):
  - H: mean building height, the mean over the columns holding a solid cell of (solid cells in the column) Δz.
  - k_c: canopy-top level, the lowest level with no solid cell; $z_c = (k_c + \frac{1}{2})\Delta z$ (so z_c/H varies by sample). Spectra, Λ_y, PDFs and W_2 are taken there. k_p = `metrics.pedestrian_level` (default 0).
  - u_{ref}: the plane mean of $\bar{u}_u$ at k_c (all cells fluid).
  - T: integral time scale of the truth. Per fluid cell and component, $x_t = u_c - \bar{u}_c$, $R(\tau) = \sum_t x_t x_{t+\tau} / \sum_t x_t^2$ (biased, not re-centred), $T_{\mathrm{cell}} = \Delta t \left(\sum_{\tau=0}^{\tau_0 - 1} R(\tau) - \frac{1}{2}\right)$ with τ_0 the first lag where $R \le 0$ (all lags if none): the trapezoid rule for $\int_0^{\tau_0 \Delta t} R\,dt$ with $R(\tau_0)$ taken as 0. T = mean over cells and components; Δt the mean output interval. $\bar{T}$ = mean of T over the samples; lead time is shown as $t/\bar{T}$ (`lead_over_T`); likewise $\bar{H}$ for $z/\bar{H}$.
- **Nondimensional per sample, then averaged.** Velocities $/u_{\mathrm{ref}}$, vorticity and divergence $\times H/u_{\mathrm{ref}}$, TKE and stresses $/u_{\mathrm{ref}}^2$, lengths $/H$, spectra $E/(u_{\mathrm{ref}}^2 H)$ against κH. Then a plain mean over the samples (one unstable sample makes a mean NaN).
- **Sources.** Every model; `persistence` (the start frame repeated: the no-skill baseline, TKE 0); `truth` (the reference curve where it has one); `truth_halves` (window statistics only, see §3).
- **Unstable rollouts.** From the first non-finite frame on, the curves are NaN; the window statistics are NaN and `stable` is false.

1 Accuracy vs lead time (`metrics.nc`; window scalars in `metrics.yaml`)

For an unstable rollout the yaml window scalars (window RMS) are NaN, while `acc_horizon_seconds` stays finite (the first NaN lead counts as a drop).

`rmse, rmse_u, rmse_v, rmse_w` [table: RMSE/u_{ref}]

$\mathrm{RMSE}(n) = \frac{1}{u_{\mathrm{ref}}}\left[\frac{1}{3|\mathcal{F}|}\sum_c \sum_{\mathbf{x}\in\mathcal{F}} (\hat{u}_c - u_c)^2\right]^{1/2}$ (pooled), per component without $\sum_c$ and the 3. *Measures:* pointwise error; grows with lead time as the rollout decorrelates. It rewards smoothing: predicting $\bar{u}$ scores the rms fluctuation, a decorrelated but realistic field $\sqrt{2}$ times that; a diverging rollout blows it up.

`rmse` (yaml), `rmse_vs_persistence`

$\left[\frac{1}{N}\sum_n \mathrm{RMSE}(n)^2\right]^{1/2}$ per sample; and its ratio to persistence's, per sample, then the mean. *Measures:* < 1: better than freezing the start frame.

`acc, acc_u, acc_v, acc_w` [table: ACC]

$\mathrm{ACC}(n) = \frac{\sum p'q'}{\sqrt{\sum p'^2 \sum q'^2}}$, $p' = \hat{u}_c - \bar{u}_c$, $q' = u_c - \bar{u}_c$, sums over $\mathbf{x} \in \mathcal{F}$ (and c for the pooled one). *Measures:* phase agreement of the fluctuations, independent of amplitude: 1 perfect, 0 no skill. Smoothing does not hurt it much; a wrong pattern does.

`acc_horizon_seconds, acc_horizon_censored` [table: $t_{\mathrm{ACC}}/\bar{T}$]

t_{ACC}: the first lead time where the pooled ACC < `metrics.acc_threshold` (default 0.5; a NaN lead counts as a drop); if it never drops, the window's last lead time and `censored` = 1 (its mean: the censored fraction). *Measures:* predictability horizon; censored values are lower bounds. The table divides the sample-mean seconds by $\bar{T}$.

`rmse_pedestrian, acc_pedestrian` [table: $\mathrm{RMSE}_{\mathrm{ped}}/u_{\mathrm{ref}}$]

The pooled RMSE and ACC on the fluid cells of level k_p only (ACC about $\bar{u}$ there); yaml: window RMS of the RMSE curve. *Measures:* accuracy where people walk, in the street canyons.

`vorticity_rmse, vorticity_rmse_x/y/z` [table: $\mathrm{RMSE}_\omega H/u_{\mathrm{ref}}$]

$(\omega_x, \omega_y, \omega_z) = \nabla \times \mathbf{u}$ by central differences; $\frac{H}{u_{\mathrm{ref}}}\big[\frac{1}{3|\mathcal{I}|}\sum_c \sum_{\mathcal{I}} (\hat{\omega}_c - \omega_c)^2\big]^{1/2}$ (pooled), per component; yaml: `vorticity_rmse` only, window RMS. *Measures:* error of the velocity gradients, i.e. of the small scales, which RMSE of $\mathbf{u}$ under-weights.

`stable`

True if the rollout is finite over the whole window (mean: all samples).

2 Physics vs lead time (`metrics.nc`, `truth` too)

`divergence_rms; divergence_ratio` (yaml) [table: D/D_{truth}]

$D(n) = \frac{H}{u_{\mathrm{ref}}}\big[\mathrm{mean}_{\mathcal{I}}(\partial_x u + \partial_y v + \partial_z w)^2\big]^{1/2}$ by central differences; ratio $= \mathrm{mean}_n \hat{D} / \mathrm{mean}_n D$ per sample. The stencil is not the solver's, so the truth's $D \neq 0$ is the reference. *Measures:* mass conservation: 1 ideal, > 1 spurious non-solenoidal noise, < 1 over-smoothing.

`tke`

$k(n) = \frac{1}{u_{\mathrm{ref}}^2}\frac{1}{2|\mathcal{F}|}\sum_c \sum_{\mathcal{F}} \big(\hat{u}_c(t_n) - \langle \hat{u}_c \rangle_t\big)^2$, about the source's *own* time mean over its finite frames (persistence: 0). *Measures:* resolved energy content against lead time: a falling curve is energy loss (smoothing), a rising one blow-up; compare with the truth's curve.

`integral_length` [table: Λ_y/H]

At k_c on the y-lines ℓ, $u'_\ell(y) = (u - \bar{u}_u)/u_{\mathrm{ref}}$ with each line's spanwise mean removed per frame (a gust over the whole line would give R a positive floor); $R(m) = \sum_\ell \sum_y u'_\ell(y)\, u'_\ell(y + m\Delta y) / \sum_\ell \sum_y u'^2_\ell$, y periodic; $\Lambda_y = \Delta y\big(\sum_{m=0}^{m_0-1} R(m) - \frac{1}{2}\big)$, m_0 the first lag with $R \leq 0$, searched up to $n_y/2$ (all lags up to $n_y/2$ if R never drops to 0): the trapezoid rule for $\int_0^{m_0\Delta y} R\, dy$ with $R(m_0)$ taken as 0; $/H$. yaml: window mean (truth too). *Measures:* size of the energetic eddies: above the truth = too coherent / smoothed, below = noisy.

`spectrum_u, spectrum_w`

Spanwise spectrum at k_c of $u' = (u - \bar{u}_u)/u_{\mathrm{ref}}$ (w'), each y-line FFT'd as is (y periodic, no window), averaged over the lines: $E(\kappa) = w_\kappa\, |\hat{u}'_\kappa / n_y|^2 / \Delta\kappa$, $\kappa = 2\pi/\lambda$ the *angular* wavenumber [rad/m], $\Delta\kappa = 2\pi/(n_y \Delta y)$, $w_\kappa = 2$ (1 at $\kappa = 0$ and the even-n_y Nyquist bin), so $\sum E\, \Delta\kappa =$ the variance along y; $\kappa = 0$ dropped; stored $/H$ against κH. In `metrics.nc`: binned on 32 shared log-spaced κH bins (mean in a bin, log-κ interpolation in an empty interior bin, NaN outside a sample's range), averaged over the samples covering the bin (dims `source`×`lead`×`kh`). *Measures:* energy per scale: a drooping tail = small scales lost (smoothing), a raised tail = noise or aliasing.

`spectra_db_{u,w}_{large,mid,near_cutoff}` (yaml) [table: $E_{u'}$ [dB] mid, near cutoff]

Per sample, with $\bar{E} = \mathrm{mean}_n E(n)$ (unbinned): $10 \log_{10}\big(\sum_{\kappa \in b} \hat{\bar{E}} / \sum_{\kappa \in b} \bar{E}\big)$, bands b by wavelength in cells $\lambda/\Delta y = 2\pi/(\kappa \Delta y)$: large $[8, \infty)$, mid $[4, 8)$, near_cutoff $[2, 4)$ (`evaluation.turbulence.SPECTRAL_BANDS`). *Measures:* 0 dB ideal; -3 dB = half the energy (smoothing), $+3$ dB = twice (noise).

3 Window statistics (`metrics.yaml`; profiles, PDFs in `metrics.nc`)

NaN for an unstable rollout. $\langle\cdot\rangle_t$: the source's own window mean; primes here are about it.

`mean_u_rmse` [table: RMSE $\bar{u}/u_{\mathrm{ref}}$]

$\big[\mathrm{mean}_{\mathcal{F}}(\langle \hat{u} \rangle_t - \langle u \rangle_t)^2\big]^{1/2}/u_{\mathrm{ref}}$. *Measures:* error of the mean flow field (wakes, recirculation).

`tke_rmse` [table: RMSE k/u_{ref}^2]

$k(\mathbf{x}) = \frac{1}{2}\sum_c \mathrm{var}_t(u_c)$ (population variance); $\big[\mathrm{mean}_{\mathcal{F}}(\hat{k} - k)^2\big]^{1/2}/u_{\mathrm{ref}}^2$. *Measures:* where turbulence is produced and how much (shear layers, wakes); a smoothed model under-predicts k.

`profile_mean_u, profile_tke, profile_uw` (nc); `profile_rmse_{mean_u,tke,uw}` [table: RMSE $-\langle u'w' \rangle/u_{\mathrm{ref}}^2$]

Per level z: the fluid-plane average (sum over the level's fluid cells / their count) of $\langle u \rangle_t/u_{\mathrm{ref}}$, of $\frac{1}{2}\sum_c \langle u'^2_c \rangle_t/u_{\mathrm{ref}}^2$ and of $-\langle u'w' \rangle_t/u_{\mathrm{ref}}^2$ (`metrics.nc` dims `source`×`z`, sample mean). RMSE: $\big[\frac{1}{n_z}\sum_z (\hat{q} - q)^2\big]^{1/2}$ over all n_z levels. *Measures:* the canopy profile; $-\langle u'w' \rangle$ is the momentum flux into the canopy.

`pdf, pdf_bin_centre` (nc)

Density of $(\hat{u}_c - \bar{u}_c)/u_{\mathrm{ref}}$ at k_c at the snapshot leads (`metrics.snapshot_leads`, -1: the last), on `max_pdf_values` / n_{samples} random cells per sample (without replacement; the same for every lead and source), pooled over the samples. Per component 60 bins over the 0.05–99.95 percentile range of the truth's values pooled over all snapshot leads and samples, the same for every source and lead; density = count/($M\, \Delta b$), M all values, so values outside fall out but count. *Measures:* turbulence intensity (width), bias (shift) and tails (intermittency) at the canopy top.

`wasserstein_u, wasserstein_v, wasserstein_w` [table: $W_2(u')/u_{\mathrm{ref}}$]

Values $(\hat{u}_c - \bar{u}_c)/u_{\mathrm{ref}}$ at k_c at $M =$ `metrics.max_pdf_values` random (lead, cell) draws per sample over the window (with replacement), the same draws for every source; $W_2 = \big[\frac{1}{99}\sum_i (Q_{\mathrm{truth}}(p_i) - Q_{\mathrm{pred}}(p_i))^2\big]^{1/2}$, Q the quantile function, $p_i = (i + \frac{1}{2})/99$, $i = 0..98$; per sample, then the mean. *Measures:* distance of the one-point distributions (bias + spread + shape) in units of u_{ref}; 0 ideal; insensitive to phase errors that ACC/RMSE punish.

`integral_length, divergence_ratio`

`integral_length`: window mean of the §2 curve; `divergence_ratio`: see §2 (a ratio of window means).

`truth_halves`

The window statistics (`mean_u_rmse` … `wasserstein_*`, incl. `spectra_db`) of the truth's first half (leads $1..\lfloor N/2 \rfloor$) against its second half. *Not a noise floor*: each half has half the frames (about $\sqrt{2}$ the sampling error of the full window), and when the inflow varies in time the halves differ by that forcing, which a model compared over the same window shares, so a model can beat it. Use it as a rough scale for long rollouts only.

4 Cost (`metrics.yaml`, models only, not per sample)

`seconds_per_step, n_parameters, speedup` [table: ms / step, Parameters (10^6), Speedup]

Inference wall time per lead step on the run's `device` (the first step of each rollout excluded: warm-up) and the parameter count (all parameters of the model) from `models.yaml`; speedup = `metrics.solver_seconds_per_step` / `seconds_per_step`, only when that is set. Time on the GPU for the cost table.

Tables (`make_tables.py`)

`table_{accuracy,physics,cost}.tex` (booktabs, to `\input`) and `.csv` (full precision). RMSE and ACC columns "$t \le h\bar{T}$" are the RMS (RMSE) or mean (ACC) of the *sample-mean* curve over the leads $t \le h\bar{T}$ for each `metrics.horizons` inside the window, and over the whole window (so the window RMSE differs slightly from the yaml `rmse`, a mean of per-sample RMS). **Bold**: the best model at the printed precision (smallest error; largest ACC, t_{ACC}, speedup; closest to 0 dB, to 1, or to the truth's Λ_y/H; smallest ms/step; parameters never bold; ties all bold; persistence and truth never). $\geq$: a lower bound (t_{ACC} with some sample censored). "–": unstable (NaN).

Figures (`visualize_comparison.py`)

In each of `figures.formats` (default PDF + PNG); the animations fall back to .gif without ffmpeg and are skipped by `figures.animation: null`.

File	Shows
`rmse`	RMSE/u_{ref} vs $t/\bar{T}$: pooled, u, v, w
`acc`	ACC vs $t/\bar{T}$: pooled, u, v, w; threshold line
`vorticity`	vorticity RMSE $\times H/u_{\mathrm{ref}}$ vs $t/\bar{T}$: pooled, x, y, z
`physics`	TKE/u_{ref}^2, divergence RMS $\times H/u_{\mathrm{ref}}$, Λ_y/H vs $t/\bar{T}$; truth as reference
`pedestrian`	RMSE/u_{ref} and ACC at k_p vs $t/\bar{T}$
`spectra`	premultiplied $\kappa E/u_{\mathrm{ref}}^2$ vs κH at k_c; rows u', w', columns the snapshot leads
`spectral_ratio`	$10\log_{10}(E_{\mathrm{model}}/E_{\mathrm{truth}})$ of u' over (κH, $t/\bar{T}$), ratio of the sample-mean spectra; ± 6 dB, contours at ratios 0.8, 1.2
`pdfs`	densities of u', v', w'/u_{ref} at k_c; columns the snapshot leads
`profiles`	plane-averaged window $\langle u \rangle_t$, TKE, $-\langle u'w' \rangle$ vs $z/\bar{H}$
`station_profiles`	the same in single columns behind the most downstream building (lee face $+\,0.5, 1, 2, 4\times$ its height)
`slices`	$\lvert\mathbf{U}\rvert/u_{\mathrm{ref}}$ at k_p; rows truth + models, columns the snapshot leads
`vertical_slices`	x–z plane through the most downstream building at `figures.vertical_lead`: u, w/u_{ref}, $\omega_y H/u_{\mathrm{ref}}$
`statistics_slices`	window $\langle u \rangle_t/u_{\mathrm{ref}}$ and TKE/u_{ref}^2 at k_p
`tke_heights`	window-averaged resolved TKE $\frac{1}{2}\sum_c \mathrm{var}_t(u_c)/u_{\mathrm{ref}}^2$ (each source about its own window mean) on horizontal slices at the heights `figures.tke_heights` (in units of H); rows truth + models
`animation_speed.mp4`	$\lvert\mathbf{U}\rvert$ at k_p vs lead time, truth and models, and model − truth (review aid)
`animation_tke_z<h>H.mp4`	instantaneous resolved TKE $\frac{1}{2}\sum_c (u_c - \langle u_c \rangle_t)^2/u_{\mathrm{ref}}^2$ about each source's own window mean, at height $h\,H$ (e.g. `z1p0H`); truth and models side by side

`station_profiles`, all `*slices` figures, `tke_heights` and the animations show trajectory `figures.sample`; persistence is left out of them; model slices share the truth's colour scale and saturate. Curves leaving the axis (an unstable model) are clipped and marked "(off scale)".